APPLICATIONS

From pilot to service

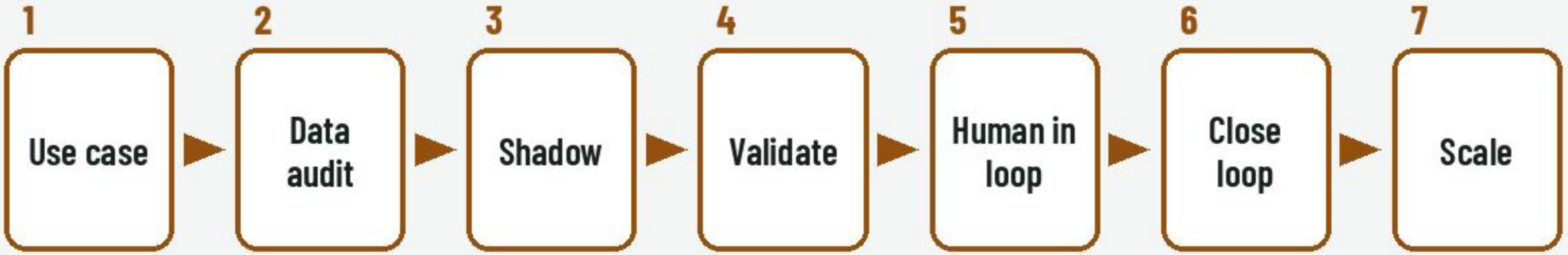
Module 12 · Safety, security, governance, implementation, research

Transport Twin Study Path

Safety and security

- A safety shell: the controller enforces minimums and fallbacks
- Cyber security: IEC 62443, NIST CSF
- Privacy: GDPR, Saudi PDPL

Rollout


1
Use case
2
Data audit
3
Shadow
4
Validate
5
Human in loop
6
Close loop
7
Scale

Governance

- Who owns data and decisions
- Who is accountable for automated actions
- Skills to run it after the pilot

Open research gaps

- Closing the loop with evidence
- Data fusion and trust
- Federation across agencies
- Human behaviour and vulnerable users

A good research question

Decision + gap + evidence

Name all three, then test on data you did not tune on